

SQL Assignment

Task:

1. Show all the customers whose creditLimit is greater than 20000

MariaDB [classicmodels]> SELECT * FROM customers WHERE creditLimit > 20000 LIMIT 10;									
	customerNumber	customerName	contactLastName	contactFirstName	phone	addressLine1	addressLine2	city	
	state	postalCode	country	salesRepEmployeeNumber	creditLimit				
	103	Atelier graphique	Schmitt	Carine	40.32.2555	54, rue Royale	NULL	Nantes	
	NULL	44000	France	1370	21000.00				
gas	112	Signal Gift Stores	King	Jean	7025551838	8489 Strong St.	NULL	Las Vegas	
	NV	83030	USA	1166	71800.00				
rne	114	Australian Collectors, Co.	Ferguson	Peter	03 9520 4555	636 St Kilda Road	Level 3	Melbourne	
	Victoria	3004	Australia	1611	117300.00				
	119	La Rochelle Gifts	Labruno	Janine	40.67.8555	67, rue des Cinquante Otages	NULL	Nantes	
	NULL	44000	France	1370	118200.00				
n	121	Baane Mini Imports	Bergulfsen	Jonas	07-98 9555	Erling Skakknes gate 78	NULL	Stavanger	
	NULL	4110	Norway	1504	81700.00				
fael	124	Mini Gifts Distributors Ltd.	Nelson	Susan	4155551450	5677 Strong St.	NULL	San Francisco	
	CA	97562	USA	1165	210500.00				
urt	128	Blauer See Auto, Co.	Keitel	Roland	+49 69 66 90 2555	Lyonerstr. 34	NULL	Frankfurt	
	NULL	68528	Germany	1504	59700.00				
ancisco	129	Mini Wheels Co.	Murphy	Julie	650555787	5557 North Pendale Street	NULL	San Francisco	
	CA	94217	USA	1165	64600.00				
	131	Land of Toys Inc.	Lee	Kwai	2125557818	897 Long Airport Avenue	NULL	NYC	
	NY	10022	USA	1323	114900.00				
	141	Euro+ Shopping Channel	Freyre	Diego	(91) 555 94 44	C/ Moralzarzal, 86	NULL	Madrid	
	NULL	28034	Spain	1370	227600.00				
10 rows in set (0.001 sec)									

2. Show the employees who report to VP Sales.

```
MariaDB [classicmodels]> SELECT * FROM employees
-> WHERE reportsTo IN (
-> SELECT employeeNumber FROM employees WHERE jobTitle = 'VP Sales'
-> );
```

employeeNumber	lastName	firstName	extension	email	officeCode	reportsTo	jobTitle
1088	Patterson	William	x4871	wpatterson@classicmodelcars.com	6	1056	Sales Manager (APAC)
1102	Bondur	Gerard	x5408	gbondur@classicmodelcars.com	4	1056	Sale Manager (EMEA)
1143	Bow	Anthony	x5428	abow@classicmodelcars.com	1	1056	Sales Manager (NA)
1621	Nishi	Mami	x101	mnishi@classicmodelcars.com	5	1056	Sales Rep

4 rows in set (0.004 sec)

3. Find all the customers who have set their state while filling the forms and Lives in USA and credit limit is between 100000 and 200000.

```
MariaDB [classicmodels]> SELECT * FROM customers
-> WHERE state IS NOT NULL
-> AND country = 'USA'
-> AND creditLimit BETWEEN 100000 AND 200000;
```

customerNumber	customerName	contactLastName	contactFirstName	phone	addressLine1	addressLine2	city	
state	postalCode	country	salesRepEmployeeNumber	creditLimit				
Y	131	Land of Toys Inc.	Lee	Kwai	2125557818	897 Long Airport Avenue	NULL	NYC
	10022	USA	1323	114900.00				
Y	151	Muscle Machine Inc	Young	Jeff	2125557413	4092 Furth Circle	Suite 400	NYC
	10022	USA	1286	138500.00				
A	157	Diecast Classics Inc.	Leong	Kelvin	2155551555	7586 Pompton St.	NULL	Allentown
	70267	USA	1216	100600.00				
A	239	Collectable Mini Designs Co.	Thompson	Valarie	7605558146	361 Furth Circle	NULL	San Diego
	91217	USA	1166	105000.00				
A	286	Marta's Replicas Co.	Hernandez	Marta	6175558555	39323 Spinnaker Dr.	NULL	Cambridge
	51247	USA	1216	123700.00				
Y	319	Mini Classics	Frick	Steve	9145554562	3758 North Pendale Street	NULL	White Plains
	24067	USA	1323	102700.00				
A	321	Corporate Gift Ideas Co.	Brown	Julie	6505551386	7734 Strong St.	NULL	San Francisco
	94217	USA	1165	105000.00				
H	363	Online Diecast Creations Co.	Young	Dorothy	6035558647	2304 Long Airport Avenue	NULL	Nashua
	62005	USA	1216	114200.00				

```
8 rows in set (0.003 sec)
```

4. Find all the employees who report to Sales Managers of all types.

```
MariaDB [classicmodels]> SELECT * FROM employees
-> WHERE reportsTo IN (
->   SELECT employeeNumber FROM employees WHERE jobTitle LIKE '%Sales Manager%'
-> );
```

employeeNumber	lastName	firstName	extension	email	officeCode	reportsTo	jobTitle
1165	Jennings	Leslie	x3291	ljennings@classicmodelcars.com	1	1143	Sales Rep
1166	Thompson	Leslie	x4065	lthompson@classicmodelcars.com	1	1143	Sales Rep
1188	Firrelli	Julie	x2173	jfirrelli@classicmodelcars.com	2	1143	Sales Rep
1216	Patterson	Steve	x4334	spatterson@classicmodelcars.com	2	1143	Sales Rep
1286	Tseng	Foon Yue	x2248	ftseng@classicmodelcars.com	3	1143	Sales Rep
1323	Vanauf	George	x4102	gvanauf@classicmodelcars.com	3	1143	Sales Rep
1611	Fixter	Andy	x101	afixter@classicmodelcars.com	6	1088	Sales Rep
1612	Marsh	Peter	x102	pmarsh@classicmodelcars.com	6	1088	Sales Rep
1619	King	Tom	x103	tking@classicmodelcars.com	6	1088	Sales Rep

9 rows in set (0.001 sec)

5. Find the average credit limit of customers of each country.

```
MariaDB [classicmodels]> SELECT country, AVG(creditLimit) AS averageCreditLimit
-> FROM customers
-> GROUP BY country;
```

country	averageCreditLimit
Australia	86060.000000
Austria	58500.000000
Belgium	51700.000000
Canada	76200.000000
Denmark	102100.000000
Finland	95266.666667
France	77691.666667
Germany	19776.923077

6. Find the total no. of orders for each date and customer. Show only dates with total number of orders greater than 10 for date and customer.

```
27 rows in set (0.004 sec)

MariaDB [classicmodels]> SELECT orderDate, customerNumber, COUNT(orderNumber) AS totalOrders
-> FROM orders
-> GROUP BY orderDate, customerNumber
-> HAVING COUNT(orderNumber) > 10;
Empty set (0.003 sec)
```

7. Find the name of the supervisor, job title of supervisor and total no. of supervisee using subquery. (With out using Join operation)

```
MariaDB [classicmodels]> SELECT
->   CONCAT(firstName, ' ', lastName) AS supervisorName,
->   jobTitle,
->   (SELECT COUNT(*) FROM employees e2 WHERE e2.reportsTo = e1.employeeNumber) AS totalSupervisees
-> FROM employees e1
-> WHERE (SELECT COUNT(*) FROM employees e2 WHERE e2.reportsTo = e1.employeeNumber) > 0;
```

supervisorName	jobTitle	totalSupervisees
Diane Murphy	President	2
Mary Patterson	VP Sales	4
William Patterson	Sales Manager (APAC)	3
Gerard Bondur	Sale Manager (EMEA)	6
Anthony Bow	Sales Manager (NA)	6
Mami Nishi	Sales Rep	1

6 rows in set (0.003 sec)

8. Find the name of the supervisor, job title of supervisor and total no. of supervisee using subquery. (With using Join operation)

```
MariaDB [classicmodels]> SELECT
->     s.firstName,
->     s.lastName,
->     s.jobTitle,
->     COUNT(e.employeeNumber) AS totalSupervisees
-> FROM employees s
-> JOIN employees e ON s.employeeNumber = e.reportsTo
-> GROUP BY s.employeeNumber, s.firstName, s.lastName, s.jobTitle;
```

firstName	lastName	jobTitle	totalSupervisees
Diane	Murphy	President	2
Mary	Patterson	VP Sales	4
William	Patterson	Sales Manager (APAC)	3
Gerard	Bondur	Sale Manager (EMEA)	6
Anthony	Bow	Sales Manager (NA)	6
Mami	Nishi	Sales Rep	1

9. Find all customers with a credit limit greater than average credit credit limit using WITH Clause.

```
MariaDB [classicmodels]> WITH AvgCredit AS (
->     SELECT AVG(creditLimit) AS avg_limit FROM customers
-> )
-> SELECT c.*
-> FROM customers c
-> JOIN AvgCredit a ON c.creditLimit > a.avg_limit;
```

customerNumber	customerName	city	state	postalCode	country	contactLastName	contactFirstName	phone	salesRepEmployeeNumber	creditLimit	addressLine1	addressLine2
112	Signal Gift Stores	Las Vegas	NV	83030	USA	King	Jean	7025551838	1166	71800.00	8489 Strong St.	NULL
114	Australian Collectors, Co.	Melbourne	Victoria	3004	Australia	Ferguson	Peter	03 9520 4555	1611	117300.00	636 St Kilda Road	Level 3
119	La Rochelle Gifts	Paris	France	75008	France	Labrun	Janine	40.67.8555	1330	116000.00	67, rue des Cinquante Otages	NULL

10. Find the rank of customer. [Customer with highest credit limit have 1 rank and Customer with lowest credit limit have highest rank]. Then, find the customer with the third highest credit limit.

```
MariaDB [classicmodels]> WITH RankedCustomers AS (
->     SELECT
->         customerName,
->         creditLimit,
->         RANK() OVER (ORDER BY creditLimit DESC) AS creditRank
->     FROM customers
-> )
-> SELECT * FROM RankedCustomers WHERE creditRank = 3;
```

customerName	creditLimit	creditRank
Vida Sport, Ltd	141300.00	3

1 row in set (0.001 sec)

11. Generate a report that shows total no. of employees working in each office.

```
MariaDB [classicmodels]> SELECT o.officeCode, o.city, COUNT(e.employeeNumber) AS totalEmployees
-> FROM offices o
-> LEFT JOIN employees e ON o.officeCode = e.officeCode
-> GROUP BY o.officeCode, o.city;
```

officeCode	city	totalEmployees
1	San Francisco	6
2	Boston	2
3	NYC	2
4	Paris	5
5	Tokyo	2
6	Sydney	4
7	London	2

7 rows in set (0.001 sec)

12. Generate a report that shows total no. of customers visited each office.

```
MariaDB [classicmodels]> SELECT o.officeCode, o.city AS officeName, COUNT(c.customerNumber) AS totalCustomers
-> FROM offices o
-> LEFT JOIN employees e ON o.officeCode = e.officeCode
-> LEFT JOIN customers c ON e.employeeNumber = c.salesRepEmployeeNumber
-> GROUP BY o.officeCode, o.city;
+-----+-----+-----+
| officeCode | officeName | totalCustomers |
+-----+-----+-----+
| 1          | San Francisco | 12          |
| 2          | Boston       | 12          |
| 3          | NYC          | 15          |
| 4          | Paris        | 29          |
| 5          | Tokyo        | 5           |
| 6          | Sydney       | 10          |
| 7          | London       | 17          |
+-----+-----+-----+
7 rows in set (0.001 sec)
```

13. Generate a report that shows total payment received by each office using payment tables and essential tables. The report should show the office name, state and country, along with total payments made.

```
MariaDB [classicmodels]> SELECT o.city AS officeName, o.state, o.country, SUM(p.amount) AS totalPayments
-> FROM offices o
-> JOIN employees e ON o.officeCode = e.officeCode
-> JOIN customers c ON e.employeeNumber = c.salesRepEmployeeNumber
-> JOIN payments p ON c.customerNumber = p.customerNumber
-> GROUP BY o.officeCode, o.city, o.state, o.country;
+-----+-----+-----+-----+
| officeName | state | country | totalPayments |
+-----+-----+-----+-----+
| San Francisco | CA    | USA    | 1337439.58 |
| Boston       | MA    | USA    | 835882.33 |
| NYC          | NY    | USA    | 1072619.47 |
| Paris        | NULL  | France | 2819168.90 |
| Tokyo        | Chiyoda-Ku | Japan | 457110.07 |
| Sydney       | NULL  | Australia | 1007292.98 |
| London       | NULL  | UK     | 1324325.90 |
+-----+-----+-----+-----+
7 rows in set (0.001 sec)
```

14. Generate a report that shows total sales(in amount) by each office using order details table and other essential tables.

```
MariaDB [classicmodels]> SELECT o.officeCode, o.city AS officeName, SUM(od.quantityOrdered * od.priceEach) AS totalSales
-> FROM offices o
-> JOIN employees e ON o.officeCode = e.officeCode
-> JOIN customers c ON e.employeeNumber = c.salesRepEmployeeNumber
-> JOIN orders ord ON c.customerNumber = ord.customerNumber
-> JOIN orderdetails od ON ord.orderNumber = od.orderNumber
-> GROUP BY o.officeCode, o.city;
+-----+-----+-----+
| officeCode | officeName | totalSales |
+-----+-----+-----+
| 1          | San Francisco | 1429063.57 |
| 2          | Boston       | 892538.62 |
| 3          | NYC          | 1157589.72 |
| 4          | Paris        | 3083761.58 |
| 5          | Tokyo        | 457110.07 |
| 6          | Sydney       | 1147176.35 |
| 7          | London       | 1436950.70 |
+-----+-----+-----+
7 rows in set (0.004 sec)
```


15. Generate a report that shows total payment pending for each office.

```
MariaDB [classicmodels]> WITH OfficeSales AS (
-> SELECT o.officeCode, o.city, SUM(od.quantityOrdered * od.priceEach) AS total_sales
-> FROM offices o
-> JOIN employees e ON o.officeCode = e.officeCode
-> JOIN customers c ON e.employeeNumber = c.salesRepEmployeeNumber
-> JOIN orders ord ON c.customerNumber = ord.customerNumber
-> JOIN orderdetails od ON ord.orderNumber = od.orderNumber
-> GROUP BY o.officeCode, o.city
-> ),
-> OfficePayments AS (
-> SELECT o.officeCode, SUM(p.amount) AS total_payments
-> FROM offices o
-> JOIN employees e ON o.officeCode = e.officeCode
-> JOIN customers c ON e.employeeNumber = c.salesRepEmployeeNumber
-> JOIN payments p ON c.customerNumber = p.customerNumber
-> GROUP BY o.officeCode
-> )
-> SELECT
-> s.officeCode,
-> s.city,
-> s.total_sales,
-> COALESCE(p.total_payments, 0) AS total_payments,
-> (s.total_sales - COALESCE(p.total_payments, 0)) AS totalPending
-> FROM OfficeSales s
-> LEFT JOIN OfficePayments p ON s.officeCode = p.officeCode;
+-----+-----+-----+-----+-----+
| officeCode | city | total_sales | total_payments | totalPending |
+-----+-----+-----+-----+-----+
| 1 | San Francisco | 1429063.57 | 1337439.58 | 91623.99 |
| 2 | Boston | 892538.62 | 835882.33 | 56656.29 |
| 3 | NYC | 1157589.72 | 1072619.47 | 84970.25 |
| 4 | Paris | 3083761.58 | 2819168.90 | 264592.68 |
| 5 | Tokyo | 457110.07 | 457110.07 | 0.00 |
| 6 | Sydney | 1147176.35 | 1007292.98 | 139883.37 |
| 7 | London | 1436950.70 | 1324325.90 | 112624.80 |
+-----+-----+-----+-----+-----+
7 rows in set (0.008 sec)
```

16. Find the creditLimit of each person, proportion of creditLimit of each person in each country. [Proportion of person in USA = creditLimit of that person / sum(creditLimit of all person in USA)]

```
MariaDB [classicmodels]> SELECT
-> customerName,
-> country,
-> creditLimit,
-> (creditLimit / SUM(creditLimit) OVER(PARTITION BY country)) AS proportion
-> FROM customers
-> WHERE creditLimit > 0;
+-----+-----+-----+-----+
| customerName | country | creditLimit | proportion |
+-----+-----+-----+-----+
| Australian Gift Network, Co | Australia | 51600.00 | 0.119916 |
| Anna's Decorations, Ltd | Australia | 107800.00 | 0.250523 |
| Souvenirs And Things Co. | Australia | 93300.00 | 0.216825 |
| Australian Collectables, Ltd | Australia | 60300.00 | 0.140135 |
| Australian Collectors, Co. | Australia | 117300.00 | 0.272601 |
| Mini Auto Werke | Austria | 45300.00 | 0.387179 |
| Salzburg Collectables | Austria | 71700.00 | 0.612821 |
| Petit Auto | Belgium | 79900.00 | 0.772727 |
| Royale Belge | Belgium | 23500.00 | 0.227273 |
| Quiltes Home Electronics Network | Canada | 48700.00 | 0.312026 |
+-----+-----+-----+-----+
```

17. Create a view showing the customer name, complete address, and their total number of orders.

```
MariaDB [classicmodels]> CREATE OR REPLACE VIEW CustomerOrdersView AS
-> SELECT
-> c.customerName,
-> CONCAT_WS(' ', c.addressLine1, c.addressLine2, c.city, c.state, c.postalCode, c.country) AS completeAddress,
-> COUNT(o.orderNumber) AS totalOrders
-> FROM customers c
-> LEFT JOIN orders o ON c.customerNumber = o.customerNumber
-> GROUP BY c.customerName, c.completeAddress;
Query OK, 0 rows affected (0.013 sec)
```

18. Update the country of a customer (use any one record).

```
MariaDB [classicmodels]> UPDATE customers  
-> SET country = 'United Kingdom'  
-> WHERE customerNumber = 103;  
Query OK, 1 row affected (0.009 sec)  
Rows matched: 1 Changed: 1 Warnings: 0
```

19. Delete all payments below 20,000.

```
MariaDB [classicmodels]> DELETE FROM payments  
-> WHERE amount < 20000;  
Query OK, 78 rows affected (0.009 sec)
```

20. Add new payments manually for an existing customer.

```
MariaDB [classicmodels]> INSERT INTO payments (customerNumber, checkNumber, paymentDate, amount)  
-> VALUES (103, 'HQ336336', '2026-05-05', 25000.00);  
Query OK, 1 row affected (0.005 sec)
```